thus P(+1), P(+3) and P(+5) appear in species such as hypophosphite, phosphite, and o-phosphate. These species are all rather stable in aqueous solutions, and have been identified through paper electrophoretic analysis. In general, it may be assumed that slow neutron irradiation of solutions of oxyanions changes the central atom to another (usually a more reduced) valence state through n,y-reactions. For example, while manganese is in the Mn(+7) state in MnO_4^-, neutron capture leads to the formation of $^{56}Mn^{2+}$ species. However, if the product valence state is unstable at ambient solution conditions, it may be immediately oxidized to a more stable higher valence state.

The *self radiolysis* of a solution may change the chemical equilibria of the solution components. For example, the α-decay of plutonium decomposes water; in a solution containing 1 mole of ^{239}Pu, ca. 0.01 mole of H_2O_2 is produced per day. This hydrogen peroxide can react with the plutonium to form a precipitate of plutonium peroxide. To avoid this precipitation, nitrite ions are added to the solution to react with the hydroxyl radicals formed by the radiolysis and to eliminate the H_2O_2.

8.8. ORGANIC COMPOUNDS

As discussed above, the chemical consequences of radiolysis depends on the physical state and the molecular composition of the irradiated material. Two properties, the dielectric constant and the electron mobility, are of great importance for the fate of the ion pairs (the radical cations and the electrons) formed on ionization.

The distance r_c (the *Onsager radius*) at which the potential energy of an ion pair corresponds to the thermal energy $\mathbf{k}T$ is, according to Onsager,

$$r_c = \mathbf{e}^2/4\pi\,\boldsymbol{\varepsilon}_0\,\varepsilon_r\,\mathbf{k}\,T \tag{8.34}$$

where $\boldsymbol{\varepsilon}_0$ and ε_r is the permittivity of free space and in the medium at distance r_c, respectively (the *Onsager equation*). The probability of an electron escaping its positive ion to become a free ion is equal to $e^{-r_c/r}$ (r is the distance travelled by the electron before it becomes thermalized) depends on the electron mobility in the medium. Most organic liquids have a lower dielectric constant than that of water and the free ion yields are, therefore, generally lower, see Table 8.4.

Table 8.4 Onsager radii, electron mobilities and free ion yields

Liquid	r_c (nm)	Mobility (cm^2/Vs)	G(free ions) (μmol/J)
Neopentane	32	55	0.09–0.11
Cyclohexane	28	0.35	0.016–0.02
Benzene	25	-	0.005–0.008
Methanol	2.3	-	0.2
Water	0.7	-	0.28

Electrons escaping their positive ions are gradually thermalized and solvated whereas electrons that recombine with their geminate (i.e. original ion pair) radical cations form excited molecules. The radical cations may react with solvent molecules. The types of reactions to be expected are e.g. proton, hydrogen atom and hydride ion transfer. Excited molecules are thus formed directly and by geminate ion recombination

$$A \sim\sim\sim\sim\rightarrow A^* \tag{8.35}$$

$$A \sim\sim\sim\sim\rightarrow A^+ + e^- \tag{8.36}$$

$$A^+ + e^- \rightarrow A^{**} \rightarrow A^* \tag{8.37}$$

The highly excited (electronically and/or vibrationally) molecule A^{**} formed by charge recombination may (within 10^{-11} s) lose part of its energy rapidly through collisions with its neighbouring molecules. The excited molecules can return to their ground state by several processes. *Singlet states* (antiparallel spins of valence electrons) de-excite within 10^{-8} s by *fluorescence*, whereas the *phosphorescence* of *triplet states* (parallel spins of the outer electrons) requires 10^{-5} to 10 s. Alternatively the excited molecules can undergo unimolecular isomerization reactions or dissociate into highly reactive radicals.

Organic molecules in general have more atoms than water and, therefore, the formation of a higher variety of products is to be expected. Both molecules smaller than the original and polymeric products are formed.

The effects of radiation on organic molecules have been shown to be strongly dependent on the molecular structure. Since the excitation energy is rapidly spread out over the whole molecule, one would expect the weakest bond to rupture, producing two radicals, provided the excitation energy exceeds the bond energy. Typical bond energies in eV are C−C $\leq$3.9, C=C (aliph) 6.4, C=C (arom) 8.4, C$\equiv$C $\leq$10, C−H 3.5−4.5, C−O 3.7 and C=O 7.7. Although C−C bonds are weaker than C−H bonds, C−H rupture predominates. For reasons as yet unknown, the C−H bond rupture is neither random nor localized to the weakest bond. Nevertheless, it is expected that compounds with unsaturated bonds are more radiation resistant than those with saturated bonds. Thus, upon irradiation with γ-rays from a ^{60}Co source, the hydrogen yield from cyclohexane is 150 times greater than that from benzene. This has been interpreted to be a result of the greater stability of the excited states of aromatic systems.

The presence of π-electrons diminishes the probability of a localization of excitation energy at a specific bond. As a result, the excitation energy is spread over the whole carbon ring and de-excitation is more likely to occur through processes such as collisional transfer rather than by dissociation. The organic compounds which are most radiation resistant contain aromatic rings (polyphenyls) and condensed ring systems (naphthalene, etc.). Their insensitivity to radiolysis has led to studies of the use of such

Table 8.5 Effects of γ-radiation on organic compounds of technical interest

Compound	Observed change at (kGy)	Useless at (kGy)	Compound	25% Reduction of desired property (kGy)
Olefins	5	10	Teflon	0.1
Silicones	5	50	Cellulose acetate	2
Mineral oils	10	100	Polyethylene	9
Alkyl aromatics	100	500	Polyvinylchloride	10
Polyphenyls	500	5000	Polystyrene	400
			Neoprene, silicon rubber	0.6
			Natural rubber	2.5

aromatic liquid hydrocarbons as cooling media in nuclear reactors. Radiation sensitivity of these compounds increases with increasing size of the aliphatic side chains but never reaches a *G*-value as high as that for a pure aliphatic compound. The primary radiolytic products of aromatic compounds are polymers. *G*-values for a large number of systems have been tabulated and can be found in the literature.

In Table 8.5 effects of γ-radiation on some organic compounds of technical interest are given. For oils the effects are mainly observed as changes in viscosity and acidity while for plastics they are associated with formation or rupture of the cross-linking. For elastomers like rubber there are changes in elasticity. For polyethylene, the following effects were observed: at 10−100 kGy the tensile strength increased; between 100 and 2000 kGy the irradiated substance became rubber and jelly-like; at 2000−5000 kGy it became hard; at >5000 kGy it became glassy but with high elasticity.

Solutes in low concentration in organic solvents react primarily with the radicals formed from the solvent. High concentration of solutes, e.g. of alkyl halides, may interfere with the charge recombination process through reactions such as

$$e^- + CH_3Cl \rightarrow \cdot CH_3 + Cl^- \tag{8.38}$$

Direct energy transfer from excited solvent molecules (A*) to solute molecules (B) to form excited solute molecules may also take place:

$$A^* + B \rightarrow B^* \tag{8.39}$$

The reprocessing of used reactor fuel elements involves solvent extraction processes with organic solvents. In these processes the solvents are subjected to high radiation fields with subsequent decomposition of the organic solvent. The design of chemical reprocessing systems must take into account any interference by the radiolytic products (Ch. 21).

Labelled compounds experience self-radiolysis induced by the radioactive decay. The extent of such radiation effects depends on the half-life, the decay energy, the specific activity of the sample, and the *G*-value for decomposition. The presence of other substances can considerably affect the amount of damage. Aromatic compounds such as

benzene (as a solvent) can serve as a protective medium to minimize radiation self-decomposition, whereas water or oxygen enhance it.

Radiation doses of 10^5 Gy can induce decomposition effects of the order of 1%. Samples whose specific activity exceed 40 GBq (1 Ci) per mol for ^{14}C or about 400 GBq (10 Ci) per mol for ^{3}H will receive a dose of this magnitude in a period of a year. Samples may be stored in benzene solution in vacuo or in deep freeze to minimize self-radiation effects and should be re-purified before use if the decomposition products are likely to affect the experiment.

The instability towards irradiation is also a problem for e.g. processes for recycling of nuclear waste by solvent extraction. Then organic molecules are used as extractant and they, together with the organic diluent decompose with time and cleaning stages have to be included. The most famous of these degradation routes is the one relating to the commonly used extraction molecule tri-butyl phosphate (TBP) which form dibutyl phosphoric acid (HDBP) with a yield of approximately $G_{HDBP} = 0.07-0.12$ μmol/J and to a lesser extent bis(1,1,3,3-tetramethylbutyl)phosphinic acid (HMBP). Both these decomposition products are extractants in their own way and disturb the extraction process by increasing the extraction of some unwanted fission and corrosion products.

8.9. EXPERIMENTAL METHODS

Radiation chemistry is characterized by the very fast generation of reactive species followed by extensive competition between recombination reactions and reactions with solutes. A complete description of a radiation chemical process requires information about the final products and the transient species.

The final products can be analyzed with standard chemical methods and much information has been gained through the use of selectively reacting scavengers.

Information about structures and identities of primary ionic species has been obtained from mass spectrometry, photoelectron and vacuum ultraviolet spectroscopy. Electron spin resonance techniques have been used extensively in the study of free radicals. Low temperature, matrixes and "spin traps" have been employed to stabilize the short lived radicals.

Typical irradiation facilities are in the case of gamma radiolysis based on either ^{60}Co or ^{137}Cs. Samples introduced in these, often intense gamma fields (up to tenths of kGy/h), will not become radioactive and the radiolysis products can be studied in normal chemical laboratories. In the case of alpha radiolysis the situation is more complicated. Alpha particles have so short range that external irradiation into a sample vial is almost impossible. Therefore it is common practice to dissolve a alpha radioactive sample into the liquid of interest. The problem is that there will be residual alpha activity present in the sample preventing analysis of degradation product in a non-radioactive laboratory and thus highly sophisticated equipment has to be dedicated and contaminated.